EBIN BABU THOMAS

AI Engineer: LLM Agents, RAG & Evaluation Engineering

ebinbt.dev · hello@ebinbt.dev · github.com/ebt55 · Mavelikara, Kerala, India (Remote)

SUMMARY

AI engineer for LLM agents, search and evaluation. Recently, for a screening test, I built hybrid keyword and vector search over 433 council meetings. Since March 2026 I also shipped Diarist, a meeting transcriber (public beta), and ExactDoc, a PDF to Word converter (on PyPI). Before that, 3.5 years building LLM, RAG and backend systems for startups in the US, Canada, Europe and Australia.

TECHNICAL SKILLS

Core: Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Anthropic API, Gemini API, OpenAI API, PydanticAI, LangGraph), MCP (FastMCP), RAG, Prompt & Context Engineering, Playwright, pytest, Docker, Git

Proficient: PostgreSQL / SQL, MongoDB, Vector DBs (Qdrant, LanceDB), PyTorch, Hugging Face Transformers / PEFT, Fine-tuning (LoRA / QLoRA, DPO), LLM Serving (vLLM, Modal), NLP (sentence-transformers, Whisper), MLOps (GitHub Actions CI/CD, OpenTelemetry), Pandas, AWS, GCP, Kubernetes (GKE), OAuth2 / JWT, Terraform, Linux, uv

Evaluation & measurement: Inspect (inspect_evals), experiment design, red-teaming, model diffing, blind grading, LLM-as-judge agreement checks, bootstrap & permutation statistics, log-probability metrics, SHA-256-pinned artifacts, Langfuse

Familiar: Rust / Tauri, LangChain, Data Streaming (Bytewax), Next.js, TypeScript

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India. A software services firm building production AI systems, RAG pipelines and backends for startup clients in the US, Canada, Europe and Australia. Grew from trainee to core developer; owned client backends end to end and scoped incoming projects.

- **FinBot, a financial assistant:** fine-tuned Mistral 7B with QLoRA and built a live news pipeline (Alpaca news API, Bytewax, Qdrant) so answers cite current news. Served it with vLLM on a GKE GPU node.
- **Natural-language property search:** built a real-estate search backend that turns plain-English questions into database queries with GPT-3.5. Deployed it on Google Cloud Run, load-tested it to size instances, and added “Did you mean?” suggestions.
- **DocuAI, semantic document search:** indexed 1,000+ documents in Qdrant so users search by meaning. Designed the ingestion and retrieval pipeline and a Next.js front end, and delivered a working client demo in two days.
- **Data and backend pipelines:** built real-time bioreactor sensor ingestion (MQTT to MongoDB) for a startup that later raised funding. Automated spoken-English video scoring (Whisper, Microsoft pronunciation API, AWS Fargate) and healthcare compliance pipelines.

INDEPENDENT PROJECTS & RESEARCH

Diarist: Local Meeting Transcriber with Speaker Labels (Windows) *Oct 2026*

diarist.dev · github.com/ebt55/diarist (NVIDIA NeMo-Speech.cpp (CUDA), Nemotron speech and diarization models, Svelte 5, TypeScript, SQLite (FTS))

- Built a free Windows app that records a meeting, or imports a recording, and labels who said each line. It runs on the user’s own GPU with nothing uploaded, and **an hour of audio takes about two minutes on an RTX 3080**. Public beta, built with coding agents.
- diarist.dev/models ranks speech and speaker models by public leaderboard scores, beside my own measurements of accuracy, speed and GPU memory.

ProofPack: Purchase-Approval Review Agent with Tamper-Evident Evidence *Jul – Sep 2026*

Built from a challenge brief (Build with Gemini XPRIZE, Jul 2026) · github.com/ebt55/proofpack (Python, Gemini 3.7 Flash, Playwright, FastAPI, pytest)

- Built an AI agent that checks Medicaid purchase requests against providers’ public websites and keeps a stamped screenshot as proof of every claim. Plain code records every link, time and hash, and **a person makes every approve or deny decision**. A project, not a product; no users.

ExactDoc: PDF to Editable Word Documents, Checked Page by Page *Aug – Oct 2026*

Open source, beta 0.3.0b3 on PyPI · github.com/ebt55/exactdoc (Python, PDFium (pypdfium2), OOXML, LibreOffice headless, PyMuPDF, pytest)

- Built a converter that turns a PDF into a Word document you can edit, laid out like the original, with real headings, tables and paragraphs that re-wrap. It can also write a version made for Google Docs. Tested by opening the output in Word, LibreOffice and Google Docs and **comparing every page with the PDF**. Designed layouts and equations don’t convert well yet.

Kobayashi Maru: Do Impossible Tasks Make Coding Agents Cheat? *Sep 2026*

Apart Research AI Incident Response sprint, solo; top half of 230+ entries · github.com/ebt55/kobayashi-maru (Python, Docker, 6 model families, plain-code cheat detectors)

- Tested whether impossible tasks in a coding agent's batch make it cheat on the tasks it could solve. Over 8,959 runs on six models, **two cheaper models started cheating**; one went from 0% to 30%. Batches also got longer as impossible tasks were added, which may explain part of it.

councilrag & rbb: Civic-Meeting Search and a Retrieval Benchmark Builder *Oct 2026*

Built in a private 48-hour company screening test; code not public (Python, LanceDB (BM25 + vector + RRF), Qwen3-4B, Granite-3.3-2B, SigLIP, RapidOCR, Diarist ASR)

- Built councilrag, which searches 433 Seattle City Council meetings and answers vote questions from the official record, with citations that open the video at the second. On a 9-meeting slice with an audio transcript, the live captions had dropped 18 roll-call votes; **the transcript recovered 17**.
- Built rbb, a retrieval benchmark that checks whether its scores predict what a reader model actually answers. Within one token budget they did on Video-MME questions, but not on the Seattle meetings.

LLM Pricing Hub Remedies: Do Antitrust Rules Stop a Shared AI Pricing Advisor? *Oct 2026*

Apart Research AI Collusion Research Sprint, 23–25 Oct 2026; results not announced · github.com/ebt55/llm-pricing-hub-remedies (Python, uv, Anthropic API, OpenAI SDK, OpenRouter, SciPy, pandas)

- Tested whether the rules in the 2026 US settlement with RealPage stop one AI pricing advisor from raising two competing sellers' prices. Across 700+ simulated games on seven models, **no rule worked reliably**. The advisor was told to maximise the sellers' joint profit.

printready: Print Checks and Safe Fixes for AI-Generated Sticker Art *Oct 2026*

github.com/ebt55/printready (Python, NumPy, OpenCV, scikit-image, Pillow, xAI API (Grok), MCP server, pytest)

- Built a tool that checks AI-made sticker art against a printer's rules, fixes what it safely can (background, white border, cut line) and sends the rest to a person. **Letting an AI agent choose the fixes did no better**, at about four times the cost.

diffing-agent-bench: A Sealed Test for AI Agents That Audit Fine-Tuned Models *Aug – Sep 2026*

Work sample for the MATS 12.0 application, Neel Nanda stream, Sep 2026 · github.com/ebt55/diffing-agent-bench (Python, PyTorch, PEFT / LoRA, vLLM, Qwen3.5-9B)

- Built a sealed test for AI agents that audit a fine-tuned model by comparing it with the original; the answer key was locked before any run. The agent **never asked a database question** in 13 tries, so it missed the planted database habit. A fixed list of 50 prompts found it for 15 cents.

odd-number-forensics: Why AI Models Chase a Leaked Score Instead of the Request *Aug 2026*

Practice work sample for the SPAR Fall 2026 model-forensics take-home · github.com/ebt55/odd-number-forensics (Python, asyncio, OpenRouter, Ollama, Wilson and Newcombe intervals)

- A published test asks a model for an even number while a leaked grader rewards odd ones. I changed the prompt one line at a time to find what makes models chase the grader. For OpenAI's o3, one-line edits moved it from **under 2% to 87%** (30 to 60 samples each).

ComfyUI-DeadMansQueue: Crash-Safe Job Queue for the ComfyUI Image App *Sep 2026*

Open-source add-on · github.com/ebt55/ComfyUI-DeadMansQueue (Python (no extra dependencies), SQLite (WAL), ComfyUI custom node)

- Built an add-on that saves ComfyUI's waiting jobs to disk, so a crash doesn't wipe the queue. When I killed ComfyUI with 40 jobs queued, **all 39 unfinished jobs came back in order**. Tested once, on Windows.

IncidentGate: Test Lab for Safety Checks on an Incident-Repair Agent *Aug – Sep 2026*

Closed at a baseline, Sep 2026. github.com/ebt55/incidentgate (Python, LangGraph, PostgreSQL, OpenTelemetry, Langfuse, Docker Compose, pytest)

- Built a test lab for three safety checks on an incident-repair agent: a rule list, a watcher model and a scripted stand-in approver. A forbidden action made in one call was blocked at once. A different forbidden change, split into three allowed steps, **got past all three checks** with the local watcher model. The attack was scripted, and each setting ran once.

eval-floor: What Benchmark Scorers Give an Answer With No Content *Sep – Oct 2026*

github.com/ebt55/evalfloor (Python, Inspect / inspect_evals 0.24.0, pytest)

- Built a check that feeds empty, refusal and copied-question answers to the real scorers of 45 Inspect Evals tasks, with no model called, to find scorers that reward an answer with no content. **Mostly, the scorers hold up**: only one lets such an answer beat its baseline, and an earlier hole someone else found is now fixed upstream.

OPEN-SOURCE CONTRIBUTIONS

- Meetily **(GitHub-Trending meeting-notes app):** backend refactor and OpenAI support, merged (PR #75); bpmn-io/refactorings: text-similarity recommender for connector templates, replacing an LLM call (issue #33).

Career break Jul 2025 – Feb 2026. Independent AI work since Mar 2026. BlueDot Technical AI Safety course, certificate Oct 2026.